

# Uplift

## AI-Powered Student Success Assistant

Early warning, similar past students and supportive action plans for Open University module teams: hybrid RAG, a multi-agent workflow, calibrated prediction models and LLM-as-judge evaluation.

Capstone project · Education domain · Open University Learning Analytics Dataset (OULAD)

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PROBLEM STATEMENT

# Advisors cannot read thousands of records in time

Students who disengage or fall behind are found too late, and dashboards say who, not why or what to do.

- Module teams hold rich data: profiles, assessment results, daily online activity, registration history, outcomes.
- Reviewing it by hand is slow and inconsistent; warning signs are spread across tables and weeks.
- By the time a student formally withdraws, it is too late to help: **75%** of withdrawals in the test presentation happened before week 15.
- Teams need early, explainable, prioritised and supportive guidance, with evidence.

**Q1** **Has a student with similar patterns struggled or withdrawn before?**

Hybrid RAG over historical cases

**Q2** **What factors are associated with poor performance or withdrawal?**

Explainable models and insights

**Q3** **Which students should be prioritised for support?**

Capacity-aware ranking and agent plans

BUSINESS NEED

# Reach the right students early, with reasons

Capacity is limited, so the top of the list must be right, and every recommendation must be defensible.

**25,204**

students still enrolled at week 15 (all modules)

**11.3%**

of them will still withdraw: the students an early warning can help

**98%**

of students contacted from the top 10% later fail or withdraw (random: 34%)

**50%**

of withdrawals are in the High tier, which flags 24% of students

- **Early**: warnings from week 5, not after the student has gone
- **Explainable**: drivers per student and what happened to similar students
- **Capacity-aware**: a ranked list for the time advisors actually have
- **Supportive and safe**: never punitive, no sensitive attributes, human review
- **Measurable**: every model and AI output evaluated, cost tracked

SOLUTION OVERVIEW

# One console, six capabilities, as of a course week

Built at weeks 5, 9, 15, 22; only students still enrolled are scored. The LLM narrates, code decides facts.

01

## Early warning

Calibrated withdrawal risk, tier, TreeSHAP drivers and outcome probabilities

02

## Who to contact first

Ranking learned for advisor capacity, with the reasons for each student

03

## Similar students (RAG)

Hybrid search over past cohorts by student or plain English, with metadata filters

04

## Support plan (agents)

LangGraph agents: performance, risk, intervention, critic; outreach draft

05

## Copilot

Questions answered by tool calls over the same computed functions, streamed and checked

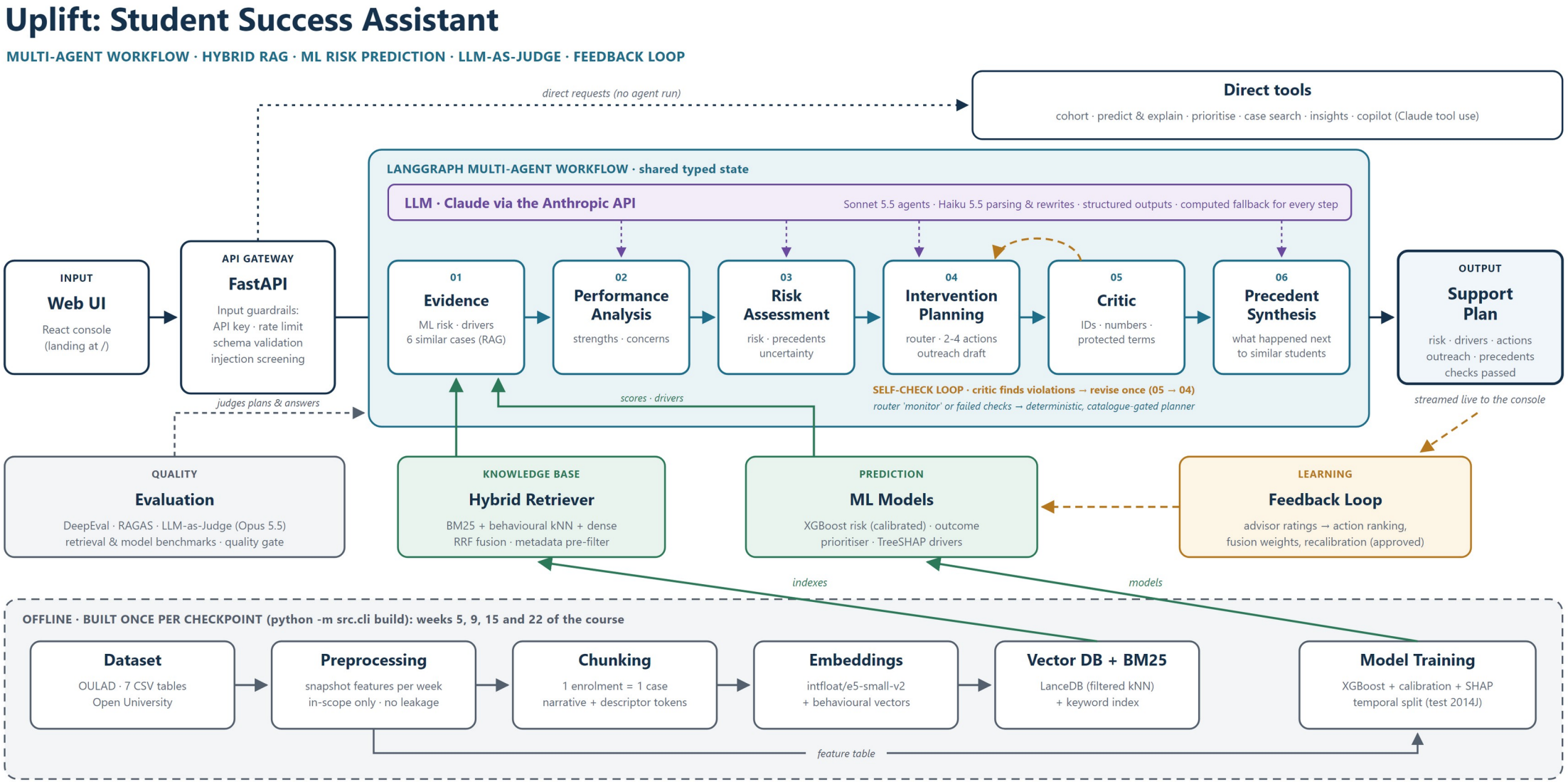
06

## Insights and evaluation

Factors, trends, fairness; DeepEval, RAGAS, judge; approved updates

ARCHITECTURE

# Offline build per checkpoint, online agents behind FastAPI

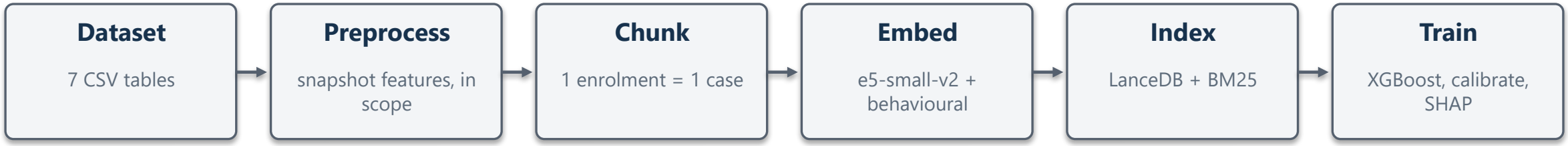


DATA FLOW

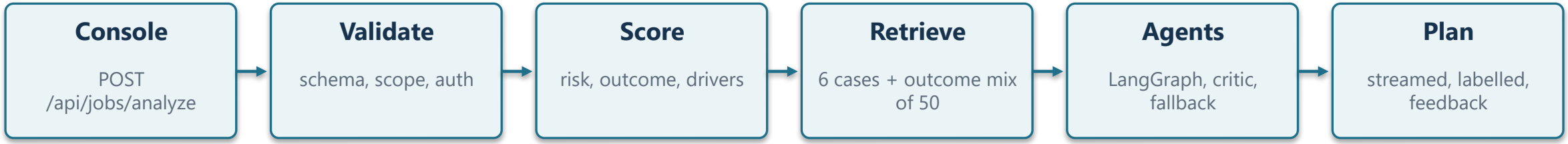
# From seven OULAD tables to a checked support plan

Only data dated before the checkpoint is used, and only students still enrolled are scored.

**OFFLINE · python -m src.cli build**




**ONLINE · draft a support plan for one student**




### The leakage rule

Scoring every enrolment, including students who had already left, gives ROC-AUC **0.749**. Scoring only students still enrolled gives **0.726**: lower, but it is the number an advisor actually gets. `date_unregistration` defines scope only; it is never a feature.

TECHNOLOGY STACK

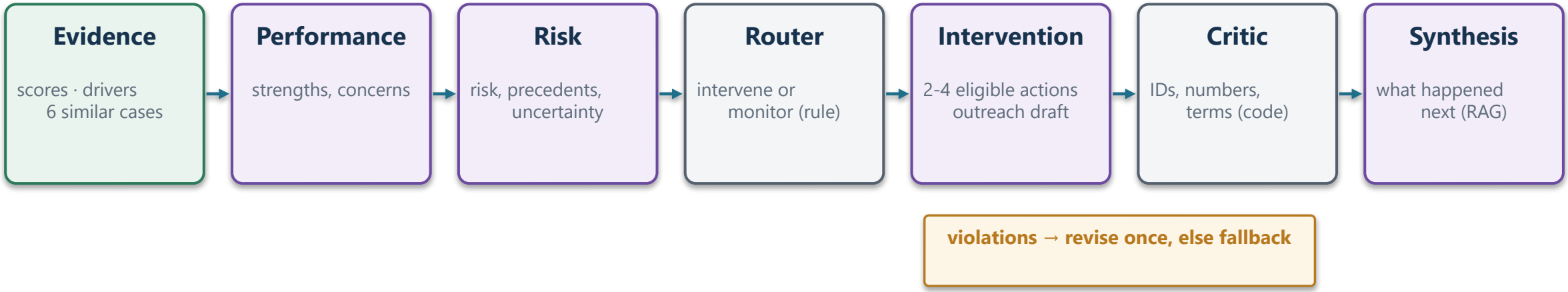
# Each choice was measured or justified against an alternative

| Layer | Choice | Why |
|---|---|---|
| **LLM** | Claude: claude-haiku-5-5 · claude-sonnet-5-5 · claude-opus-5-5 (judge) | Measured per task: small model parses as well; main model writes most faithfully; different judge |
| **Embeddings** | intfloat/e5-small-v2 | Best of five models on precedent quality at MiniLM latency |
| **Vector DB** | LanceDB (embedded) | No server; metadata pre-filter; exact kNN; parity with brute force |
| **Agents** | LangGraph state graph | Explicit router, critic and revision; parity with plain Python |
| **Prediction** | XGBoost + calibration + TreeSHAP | ROC-AUC 0.726 vs 0.703 logistic; exact explanations |
| **Retrieval** | BM25 + behavioural kNN + dense, RRF | Numbers need numeric similarity; text needs keywords and meaning |
| **API** | FastAPI + Uvicorn, pydantic | Typed validation, OpenAPI docs, streaming, background jobs |
| **UI** | React 19 + TypeScript (Vite) | Beyond Streamlit: URLs, virtual tables, streaming; served by FastAPI |
| **Evaluation** | DeepEval · RAGAS · LLM judge · pytest | Each layer catches a different failure |

AGENT WORKFLOW

# Agents on a shared state; code checks every draft

Agents cite numbered evidence; the critic rejects unknown IDs, ineligible actions, ungrounded numbers and unsafe words.




**Purple** = Claude drafts (structured JSON output)

**Grey** = code only, deterministic

**Green** = ML models and hybrid retriever

Every LLM step has a computed fallback: the workflow runs with the LLM switched off.

**100%**

plans pass every guardrail

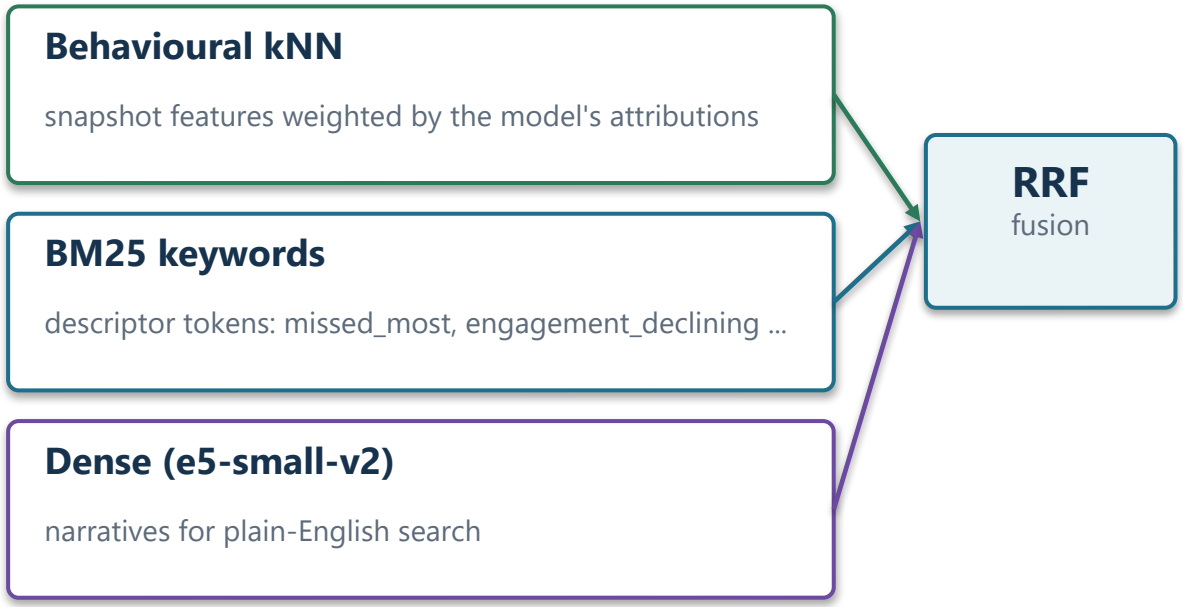
**96%**

judge pass rate (claude-opus-5-5)

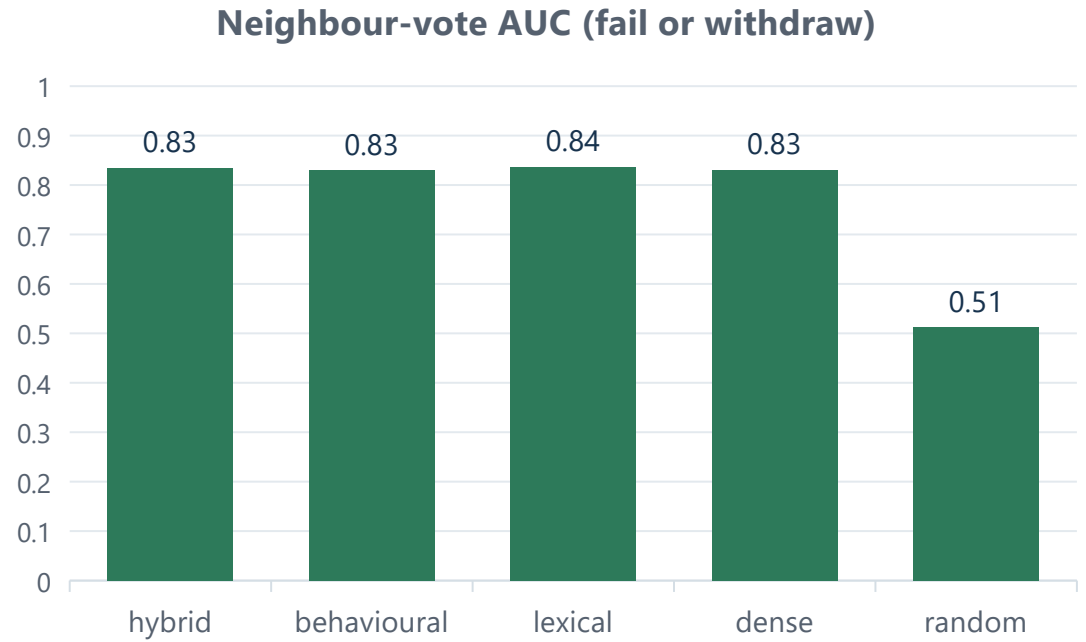
HYBRID RAG

# Three channels, fused with RRF, filtered in LanceDB

Earlier presentations only, never the student's own rows; one document per enrolment; numbers banded, not embedded.




- Metadata filters: course, region, education, age band, credits, previous attempts, outcome
- Plain English understood as filters + behaviours (Claude Haiku, rule fallback)
- Context: 6 closest stories; outcome mix of the 50 nearest




Do the retrieved students' real outcomes predict this student's? 600 test students vs earlier cohorts (not circular).

**0.862**

precision@10, plain-English case search (AI parser)

PREDICTION MODELS

# Calibrated risk that holds on a later, untouched cohort

Trained on 2013B + 2013J, tuned on 2014B, tested once on 2014J.

**0.726**

ROC-AUC at week 15 (95% CI 0.708 to 0.740)

**0.009**

calibration error: a 20% risk means about 20%

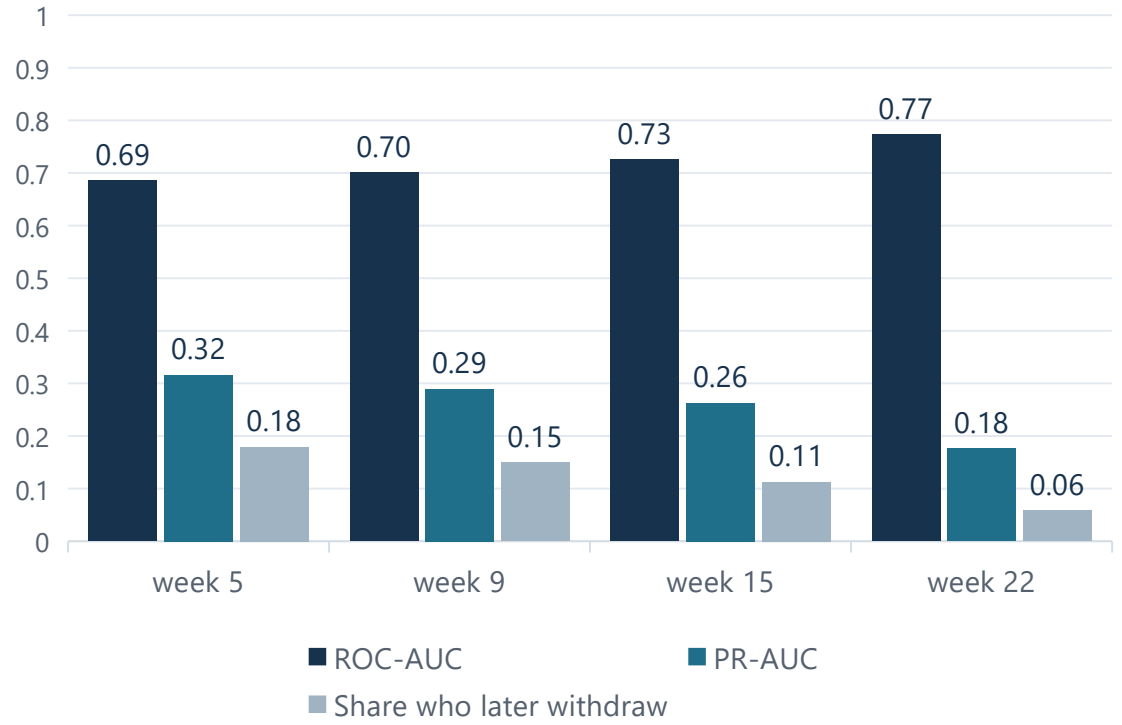
**0.703**

logistic regression baseline

**0.263**

PR-AUC vs base rate 0.113

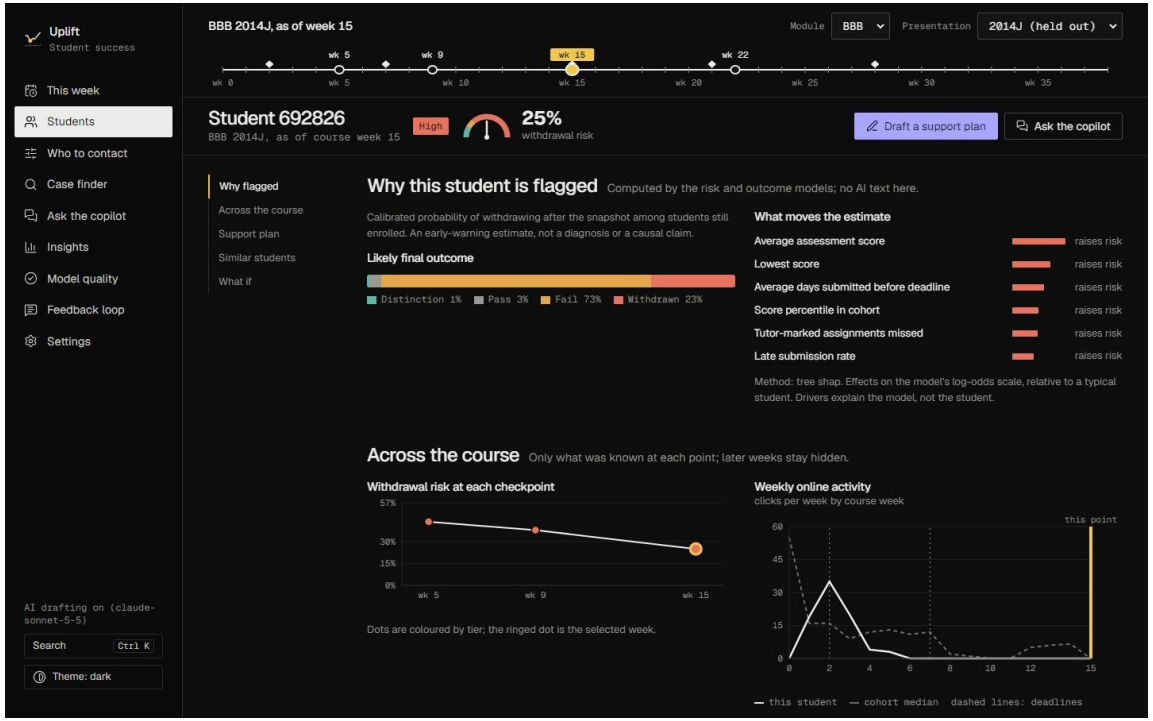
| Tier | Share of students | Observed withdrawal rate | Share of withdrawals |
|---|---|---|---|
| **High** | 24% | 23.6% | 50% |
| **Medium** | 28% | 11.7% | 29% |
| **Low** | 48% | 4.9% | 21% |



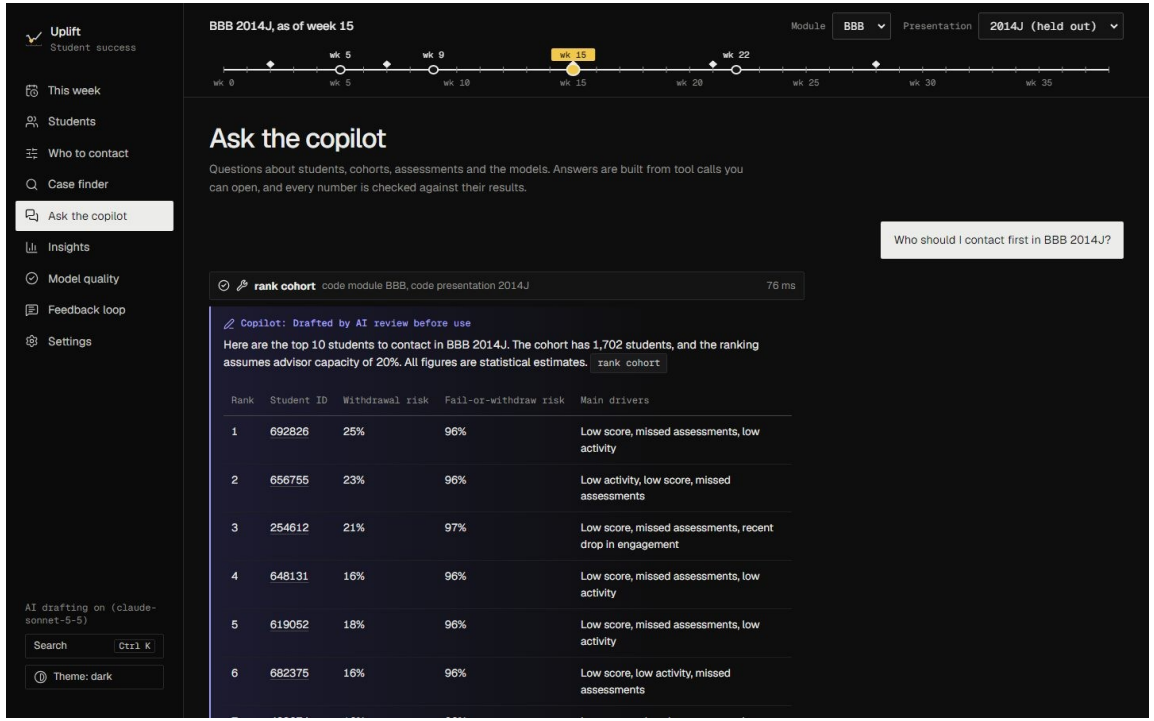

Outcome model macro-F1 0.485 (logistic 0.460); P(fail or withdraw) ROC-AUC 0.852. Ranking precision at 10%: 98%.

DEMO

# Live: week, student, support plan, case search, copilot




Student case file: why flagged, risk across the course, support plan




Copilot: tool calls, streamed answer, every number checked

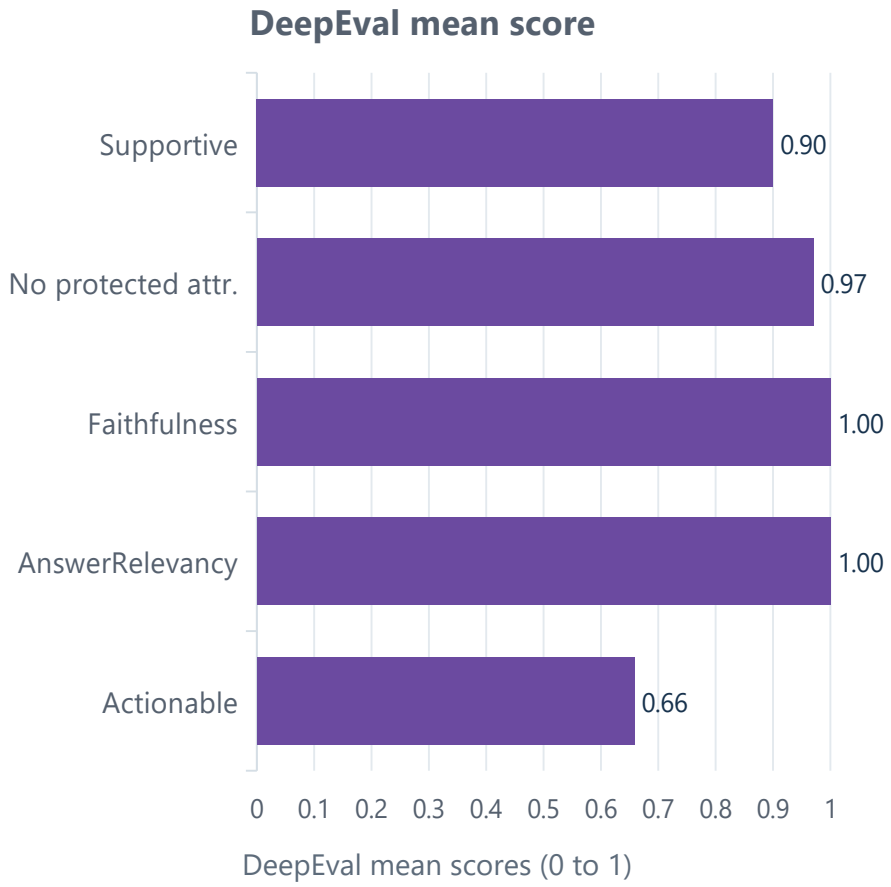
1. uvicorn src.main:app --reload → http://127.0.0.1:8000
2. This week: course clock, week 5 → 15, who to reach first
3. Student: drivers, risk trail, Draft a support plan (agent steps), rewrite as SMS
4. Case finder: plain-English search
5. Copilot: 'Who should I contact first in BBB 2014J?'

EVALUATION RESULTS

# Five layers of evaluation, each catching what the others miss

AI evaluation: generators claude-sonnet-5-5 / claude-haiku-5-5; judge claude-opus-5-5; release gate passed.

| Layer | Measure | Result |
|---|---|---|
| **1 Code checks** | plans passing every guardrail | 100% |
| **2 LLM-as-judge** | judge pass rate · AI plan preferred (order-swapped A/B) | 96% · 89% |
| **3 DeepEval** | faithfulness · answer relevancy · actionable (pass rate) | 100% · 100% · 75% |
| **4 RAGAS** | faithfulness of precedent synthesis (hallucination rate) | 0.964 (3.6%) |
| **5 Copilot** | golden questions correct · adversarial prompts handled safely | 100% · 88% |
| **Retrieval** | neighbour-vote AUC hybrid vs random | 0.835 vs 0.512 |
| **Tests** | pytest suite: no network, no LLM | 139 tests |



RESPONSIBLE AI

# Supportive by design, checked in code, fair by audit

### Input guardrails

API key and admin key
Rate limit, request-size limit
Strict schemas and vocabularies
Prompt-injection screening

### Output guardrails

IDs must exist, numbers must be in the evidence
Only eligible, supportive actions
No protected, punitive, clinical or stigma terms
Revise once, else computed fallback

### Human in the loop

AI text labelled; advisor edits every message
Never for grading, admissions or sanctions
Feedback changes nothing until approved

- **Sensitive attributes** (gender, region, deprivation, age, education, disability) are never model inputs and never sent to the LLM; adding them would change ROC-AUC by only 0.002.
- **Fairness audit**: each group gap is tested against the gap random labels would give at the same group sizes (permutation test, overall and within module).
- Region and deprivation gaps are within chance. The disability gap (0.23) is real and reported, with a per-group threshold review for the equity office, not applied automatically.

CHALLENGES AND LEARNINGS

# What was hard, and what changed because of it

| Challenge | What we did | Learning |
|---|---|---|
| **Inflated accuracy (0.90 AUC) from scoring students who had left** | Score in-scope students only; temporal split | Evaluate the decision the user actually faces |
| **Numbers in text made keyword and dense search random** | Band numbers into descriptors; numeric similarity in its own channel | Hybrid retrieval must match the data type |
| **LLM drafts invented numbers and IDs** | Evidence IDs, code checks, one revision, fallback | Let the LLM narrate and select; code decides facts |
| **Self-grading and position bias** | Different judge model, computed verdict, order-swapped A/B, human review | Evaluate the evaluator |
| **Validation gains that did not hold on test** | Rejected the feature set instead of tuning on test | Keep the test set untouched |
| **RAGAS dependencies broke the pinned service** | RAGAS in its own environment | Isolate evaluation tooling |



# Thank you

Questions, or a live look at any part of Uplift.

Run it: uvicorn src.main:app --reload → http://127.0.0.1:8000

Architecture_Diagram.pdf · Design_Document.pdf · API_Documentation.pdf · Evaluation_Report.pdf · Dataset_Details.pdf

API reference live at /api/docs · model card: docs/MODEL_CARD.md